Smart Home Management System (Smart Home+)

Database Design Documentation

Version 1.0

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1.Introduction

1.1 Purpose

The purpose of this document is to present the database design of the Smart Home Management System. It describes the logical structure of the database, including entities, attributes, relationships, normalization, keys, constraints, and SQL implementation. The document serves as a technical reference for developers, database administrators, testers, and future maintainers to ensure that the database is designed in a consistent, secure, and scalable manner.

1.2 Scope

The Smart Home Management System database is designed to store and manage information related to users, rooms, smart devices, permissions, automation rules, notifications, activity logs, and energy consumption. The database supports core system functionalities such as user authentication, device management, room organization, access control, automation, activity tracking, and energy monitoring. The design focuses on maintaining data integrity, minimizing redundancy through normalization, and supporting future system expansion.

1.3 Objectives

The primary objectives of the database are:

- Store user information securely.
- Manage rooms and smart devices efficiently.
- Support role-based access control.
- Maintain automation rules for smart home operations.
- Record user activities and system events.
- Manage notifications for users.
- Monitor and store energy consumption.
- Ensure data integrity through normalization and relational design.
- Provide a scalable database architecture for future enhancements.

1.4 Intended Audience

This document is intended for:

- Software Developers
- Database Developers
- Database Administrators (DBA)
- Software Testers
- Project Maintainers
- Students and Researchers studying the project

1.5 Technology Stack

Component	Technology
Database	SQLite 3
Backend Framework	ASP.NET Core Web API (.NET 8)
Programming Language	C#
Mobile Application	Android (Future Development)
Integrated Development Environment (IDE)	Visual Studio 2022
Version Control System	Git
Repository Hosting	GitHub

2. Database Overview

The Smart Home Management System database is designed to serve as a structured, reliable, and central repository for storing and managing all core application data. It provides a robust architecture to handle information across various functional domains, including:

- **User & Access Control:** Users, Roles, Permissions, and Notifications.
- **Environment & Hardware:** Rooms, Smart Devices, and Activity Logs.
- **Automation & Analytics:** Automation Rules and Energy Consumption Tracking.

The database follows the Relational Database Management System (RDBMS) model. Data is organized into normalized, interconnected tables using Primary Keys and Foreign Keys to enforce referential integrity, maintain data consistency, and minimize redundancy.

For the current development and simulation phase, SQLite 3 is selected as the database engine. It offers a lightweight, serverless, and zero-configuration environment that seamlessly integrates with Entity Framework Core 8. Furthermore, the database schema is designed with high adaptability, allowing effortless future migration to Microsoft SQL Server for enterprise-scale deployment with minimal architectural changes.

2.2 Database Environment

Category	Specification
Database Name	SmartHomeDB
Database Type	Relational Database Management System (RDBMS)
Planned DBMS	SQLite 3 *(Subject to implementation)*

Database File	SmartHomeDB.db *(Will be generated during implementation)*
Object-Relational Mapper (ORM)	Entity Framework Core 8
Backend Framework	ASP.NET Core Web API (.NET 8)
Programming Language	C#

2.3 Database Architecture

The Smart Home Management System follows a three-tier architecture, where the database functions as the data storage layer. The mobile application communicates with the ASP.NET Core Web API, which processes client requests and interacts with the SQLite database through Entity Framework Core. This architecture separates the presentation layer, business logic layer, and data access layer, resulting in improved maintainability, scalability, and security.

The overall database architecture can be represented as follows:

Mobile Application

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ASP.NET Core Web API

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Entity Framework Core 8

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SQLite 3 Database (SmartHomeDB)

2.4 Database Components

The database consists of multiple relational tables, each responsible for storing a specific category of information.

Table Name	Purpose

Users	Stores user account information and authentication details.
Rooms	Stores information about rooms within the smart home.
Devices	Stores information about smart devices installed in each room.
Permissions	Manages user access permissions for smart devices.
AutomationRules	Stores automation rules for controlling smart devices automatically.
Notifications	Stores notifications generated by the system for users.
ActivityLogs	Records of user activities and device-related events.
EnergyConsumptions	Stores energy usage information for smart devices.

2.5 Database Features

The Smart Home Management System database provides the following features:

- Centralized storage for all smart home data.
- Relational database design to maintain data consistency and integrity.
- Secure storage of user information and role-based access control.
- Efficient management of rooms, devices, and user permissions.
- Support for automation rules and smart device operations.
- Activity logging for monitoring user and system actions.
- Notification management for important system events.
- Energy consumption monitoring and reporting.
- Scalable database structure that supports future migration to Microsoft SQL Server.

3. Database Design

3.1 Entity Identification

Entity identification is the process of determining the major objects or concepts that need to be stored in the database. Each entity represents a real-world object or activity within the Smart Home Management System. These entities are designed to organize data efficiently while maintaining consistency and supporting the functional requirements of the system.

Based on the system requirements, the Smart Home Management System consists of the following entities:

Identified Entities

Entity Name	Description
Users	Stores user account information, authentication credentials, and role details.
Rooms	Stores information about the rooms available in the smart home.
Devices	Stores details of smart devices installed in different rooms.
Permissions	Stores user access permissions for controlling specific smart devices.
AutomationRules	Stores automation rules used to control smart devices automatically.
Notifications	Stores notifications generated by the system for users.
ActivityLogs	Records of user activities and device-related events.
EnergyConsumptions	Stores energy usage records of smart devices for monitoring and reporting purposes.

3.2 Attribute Identification

Attribute identification is the process of determining the data elements that describe each entity in the database. Attributes define the properties or characteristics of an entity and store the information required to support the functionalities of the Smart Home Management System. Each entity contains a set of attributes that uniquely identify records and maintain the necessary information for efficient data storage, retrieval, and management.

The attributes identified for each entity are listed below.

3.2.1 Users Attribute

Attribute	Description
U_Id	Unique identifier of the user.
U_Name	Full name of the user.
U_Mail	Email address used for login and communication.
U_Pass	Encrypted password of the user.
U_Role	Role assigned to the user (Homeowner, Family Member, Guest, or Technician).
U_Acc_Status	Indicates whether the user account is Active or Inactive.

3.2.2 UserPhoneNumbers Attributes

Attribute	Description
U_Id	References to the associated user.
U_Number	Contact the number of the user.

3.2.3 Rooms Attributes

Attribute	Description
R_Id	Unique identifier of the room.
R_Name	Name of the room.
R_Type	Type or category of the room (e.g., Bedroom, Kitchen, Living Room).

3.2.4 Devices Attributes

Attribute	Description
D_Id	Unique identifier of the smart device.
D_Name	Name of the smart device.
D_Type	Type or category of the device (e.g., Light, Fan, Television, Appliance).
D_Status	Current operational status of the device (e.g., Active, Inactive, Faulty, Maintenance).
D_PowerState	Indicates whether the device is ON or OFF.
D_InstallationDate	Date when the device is installed.
R_Id	References to the room where the device is installed.

3.2.5 Permissions Attributes

Attribute	Description
P_Id	Unique identifier of the permission record.
P_Status	Indicates whether the permission is Active or Inactive.
P_AccessType	Specifies the level of access granted to the user (e.g., Full Access, Control, Read Only).
P_GrantedDate	Date when the permission was granted.
P_ExpireDate	Date when permission expires.
U_Id	References the user receiving permission.
D_Id	References the smart device associated with the permission.

3.2.6 UserRooms Attributes

Attribute	Description
U_Id	References the user assigned to a room.
R_Id	References the assigned room.

3.2.7 AutomationRules Attributes

Attribute	Description
A_Id	Unique identifier of the automation rule.
A_Name	Name of the automation rule.
A_TriggerCondition	Condition that activates the automation rule.
A_Action	Action performed when the rule is triggered.
A_IsActive	Indicates whether the automation rule is active.
D_Id	References the smart device associated with the automation rule.

3.2.8 Notifications Attributes

Attribute	Description
N_Id	Unique identifier of the notification.
N_Message	Detailed notification message sent to the user.
N_DateTime	Date and time when the notification was generated.
N_Type	Type or category of the notification.
N_IsRead	Indicates whether the notification has been read.
U_Id	References the user receiving the notification.

3.2.9 Activity log Attributes

Attribute	Description
AL_Id	Unique identifier of the activity log.
AL_Action	Type of activity performed by the user.
AL_Description	Detailed description of the activity.
AL_DateTime	Date and time when the activity occurred.
U_Id	References the user who performed the activity.
D_Id	References the smart device associated with the automation rule.

3.2.10 EnergyConsumption Attributes

Attribute	Description
E_Id	Unique identifier of the energy consumption record.
E_PowerUsed	Amount of energy consumed by the device.

E_DateTime	Date and time when the energy consumption was recorded.
D_Id	References the smart device whose energy consumption was recorded.

3.3 Relationship Identification

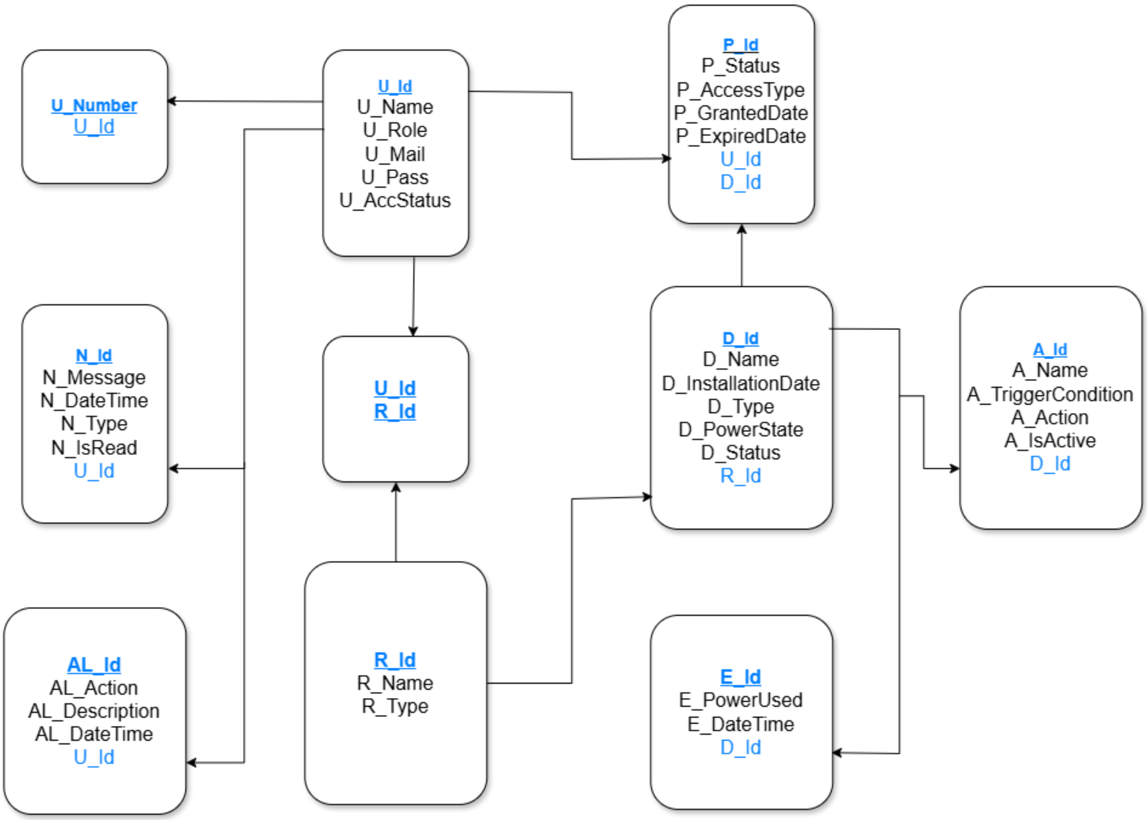
Relationship identification is the process of defining how the entities in the database are associated with one another. These relationships establish connections between different tables using primary keys and foreign keys, ensuring data consistency, integrity, and efficient retrieval. The Smart Home Management System database primarily follows **one-to-many** relationships, where one record in a parent table can be associated with multiple records in a child table.

The relationships identified in the Smart Home Management System database are described below.

Parent Entity	Child Entity	Relationship	Description
Users	Permissions	One-to-Many	One user can have multiple device permissions, while each permission belongs to one user.
Devices	Permissions	One-to-Many	One smart device can have multiple permission records, while each permission is associated with one smart device.
Users	UserPhoneNumber	One-to-Many	One user can have multiple phone numbers, while each phone number belongs to one user.
Users	UserRooms	One-to-Many	One user can be assigned to multiple rooms, while each user-room assignment belongs to one user.
Rooms	UserRooms	One-to-Many	One room can be assigned to multiple users, while each user-room assignment belongs to one room.
Rooms	Devices	One-to-Many	One room can contain multiple smart devices, while each device belongs to one room.
Users	Notifications	One-to-Many	One user can receive multiple notifications, while each notification belongs to one user.

Devices	AutomationRules	One-to-Many	One smart device can have multiple automation rules, while each automation rule is associated with one smart device.
Users	ActivityLogs	One-to-Many	One user can generate multiple activity logs, while each activity log belongs to one user.
Devices	EnergyConsumptions	One-to-Many	One device can have multiple energy consumption records, while each record belongs to one device.

3.4 Relational Schema Diagram



3.5 Primary Key & Foreign Key Specification

Table	Primary Key (PK)	Foreign Key (FK)
Users	U_Id	
UserPhoneNumbers	(U_Id, U_Number)	U_Id → Users(U_Id)
Rooms	R_Id	
Devices	D_Id	R_Id → Rooms(R_Id)
Permissions	P_Id	U_Id → Users(U_Id), D_Id → Devices(D_Id)
UserRooms	(U_Id, R_Id)	U_Id → Users(U_Id), R_Id → Rooms(R_Id)
AutomationRules	A_Id	D_Id → Devices(D_Id)
Notifications	N_Id	U_Id → Users(U_Id)
ActivityLogs	AL_Id	U_Id → Users(U_Id), D_Id → Devices(D_Id)
EnergyConsumptions	E_Id	D_Id → Devices(D_Id)

3.6 Normalization

Relation Set: Generate (one to many)

UNF: U_Id, U_Mail,{ U_Number}, U_Name, U_Role, U_Pass, U_Acc_Status, AL_Id, Al_Action, AL_Description, AL_DateTime.

1NF: U_Number are multivalued attributes.

1. U_Number, U_Id.

2NF:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.

2. AL_Id, Al_Action, AL_Description, AL_DateTime, U_Id.

3. U_Number, U_Id.

3NF:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.

2. AL_Id, Al_Action, AL_Description, AL_DateTime, U_Id.

3. U_Number, U_Id.

Relation Set: Receive permission (one to many)

UNF: U_Id, U_Mail,{ U_Number}, U_Name, U_Role, U_Pass, U_Acc_Status, P_Id , P_Status, P_AccessType, P_GrantedDate, P_ExpireDate

1NF: U_*Number are multivalued attributes.*

1. U_Number, U_Id.

2NF:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.

2. P_Id , P_Status, P_AccessType, P_GrantedDate, P_ExpireDate, U_Id.

3. U_Number, U_Id.

3NF:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.

2. P_Id , P_Status, P_AccessType, P_GrantedDate, P_ExpireDate, U_Id.

3. U_Number, U_Id.

Relation Set: Granted Access to (One to many)

UNF:

P_Id , P_Status, P_AccessType, P_GrantedDate, P_ExpireDate, D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status.

1NF:

There is no multivalued attribute.

2NF:

1. P_Id , P_Status, P_AccessType, P_GrantedDate, P_ExpireDate, D_Id.

2. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status .

3NF:

1. P_Id , P_Status, P_AccessType, P_GrantedDate, P_ExpireDate, D_Id.

2. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status .

Relation Set: Manage (Many to many)

UNF: U_Id, U_Mail,{ U_Number}, U_Name, U_Role, U_Pass, U_Acc_Status, R_Id, R_Type, R_Name.

1NF: U_Number are multivalued attributes.

1. U_Number, U_Id.

2NF:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.

2. R_Id, R_Type, R_Name.

3. U_Number, U_Id.

4. U_Id, R_Id.

3NF:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.

2. R_Id, R_Type, R_Name.

3. U_Number, U_Id.

4. U_Id, R_Id.

Relation Set: Contains (One to Many)

UNF: R_Id, R_Type, R_Name, D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status.

1NF: There is no multivalued attribute.

2NF:

1. R_Id, R_Type, R_Name.
2. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status, R_Id.

3NF:

1. R_Id, R_Type, R_Name.

2. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status, R_Id.

Relation Set: Receives (One to Many)

UNF: U_Id, U_Mail,{ U_Number}, U_Name, U_Role, U_Pass, U_Acc_Status, N_Id , N_Message, N_DateTime, N_Type, N_IsRead.

1NF: U_Number is a multivalued attribute.

1. U_Number, U_Id.

2NF:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.

2. N_Id , N_Message, N_DateTime, N_Type, N_IsRead,U_Id.

3. U_Number, U_Id.

3NF:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.

2. N_Id , N_Message, N_DateTime, N_Type, N_IsRead,U_Id.

3. U_Number, U_Id.

Relation Set: Uses (One to Many)

UNF: D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status, A_Id,A_Action, A_Name,A_TriggerCondition, A_IsActive.

1NF: There is no multivalued attribute.

2NF:

1. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status.
2. A_Id,A_Action, A_Name,A_TriggerCondition, A_IsActive, D_Id.

3NF:

1. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status.
2. A_Id,A_Action, A_Name,A_TriggerCondition, A_IsActive, D_Id.

Relation Set: Consume (One to Many)

UNF: D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status, E_Id, E-PowerUsed, E_DateTime

1NF: There is no multivalued attribute.

2NF:

1. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status.

2. E_Id, E-PowerUsed, E_DateTime,D_Id.

3NF:

1. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status.

2. E_Id, E-PowerUsed, E_DateTime,D_Id.

Generalization:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.
2. AL_Id, Al_Action, AL_Description, AL_DateTime, U_Id.
3. U_Number, U_Id.
4. ~~U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.~~
5. P_Id , P_Status, P_AccessType, P_GrantedDate, P_ExpireDate, U_Id.
6. ~~U_Number, U_Id.~~
7. P_Id , P_Status, P_AccessType, P_GrantedDate, P_ExpireDate, D_Id.
8. ~~D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status.~~
9. ~~U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.~~
10. R_Id, R_Type, R_Name.
11. ~~U_Number, U_Id.~~
12. ~~R_Id, R_Type, R_Name.~~
13. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status, R_Id.

14. ~~U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.~~
15. N_Id , N_Message, N_DateTime, N_Type, N_IsRead,U_Id.
16. ~~U_Number, U_Id.~~
17. ~~D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status.~~
18. A_Id,A_Action, A_Name,A_TriggerCondition, A_IsActive, D_Id.
19. ~~D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status.~~
20. E_Id, E-PowerUsed, E_DateTime,D_Id.
21. U_Id, R_Id.

Final Table:

1. U_Id, U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status.
2. AL_Id, Al_Action, AL_Description, AL_DateTime, U_Id.
3. U_Number, U_Id.
4. P_Id , P_Status, P_AccessType, P_GrantedDate, P_ExpireDate, U_Id, D_Id.
5. D_Id, D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status, R_Id.
6. R_Id, R_Type, R_Name.
7. N_Id , N_Message, N_DateTime, N_Type, N_IsRead,U_Id.
8. A_Id,A_Action, A_Name,A_TriggerCondition, A_IsActive, D_Id.
9. E_Id, E_PowerUsed, E_DateTime,D_Id.
10. U_Id, R_Id.

3.7 Data Dictionary

The data dictionary provides a detailed description of the attributes used in the Smart Home Management System database. It specifies the attribute name, data type, key constraint, nullability, and purpose of each attribute.

3.7.1 Users

Attribute	Data Type	Key	Nullable	Description
U_Id	INTEGER	PK	No	Unique identifier of the user.
U_Mail	TEXT	—	No	Email address of the user.
U_Name	TEXT	—	No	Full name of the user.
U_Role	TEXT	—	No	Role assigned to the user.
U_Pass	TEXT	—	No	Encrypted password of the user.

U_Acc_Status	TEXT	—	No	Indicates whether the user account is active or inactive.

3.7.2 UserPhoneNumbers

Attribute	Data Type	Key	Nullable	Description
U_Number	TEXT	PK,FK	No	Contact number of the user.
U_Id	INTEGER	PK,FK	No	References the associated user.

Foreign Key: `U_Id → Users(U_Id)`

3.7.3 Rooms

Attribute	Data Type	Key	Nullable	Description
R_Id	INTEGER	PK	No	Unique identifier of the room.
R_Type	TEXT	—	No	Type or category of the room.
R_Name	TEXT	—	No	Name of the room.

3.7.4 Devices

Attribute	Data Type	Key	Nullable	Description
D_Id	INTEGER	PK	No	Unique identifier of the smart device.
D_Name	TEXT	—	No	Name of the smart device.
D_InstallationDate	TEXT	—	No	Date when the device was installed.
D_Type	TEXT	—	No	Type or category of the device.
D_PowerState	TEXT	—	No	Indicates whether the device is ON or OFF.
D_Status	TEXT	—	No	Current operational status of the device.
R_Id	INTEGER	FK	No	References the room where the device is installed.

Foreign Key: `R_Id → Rooms(R_Id)`

3.7.5 Permissions

Attribute	Data Type	Key	Nullable	Description
P_Id	INTEGER	PK	No	Unique identifier of the permission record.
P_Status	TEXT	—	No	Indicates whether the permission is active or inactive.
P_AccessType	TEXT	—	No	Specifies the type of access granted to the user.
P_GrantedDate	TEXT	—	No	Date when the permission was granted.
P_ExpireDate	TEXT	—	Yes	Date when the permission expires.
U_Id	INTEGER	FK	No	References the user receiving the permission.
D_Id	INTEGER	FK	No	References the smart device associated with the permission.

Foreign Keys:

- `U_Id → Users(U_Id)`
- `D_Id → Devices(D_Id)`

3.7.6 UserRooms

Attribute	Data Type	Key	Nullable	Description
U_Id	INTEGER	PK, FK	No	References the user assigned to the room.
R_Id	INTEGER	PK, FK	No	References the assigned room.

Composite Primary Key: `(U_Id, R_Id)`

Foreign Keys:

- `U_Id → Users(U_Id)`
- `R_Id → Rooms(R_Id)`

3.7.7 AutomationRules

Attribute	Data Type	Key	Nullable	Description
A_Id	INTEGER	PK	No	Unique identifier of the automation rule.
A_Action	TEXT	—	No	Action performed when the rule is triggered.
A_Name	TEXT	—	No	Name of the automation rule.
A_TriggerCondition	TEXT	—	No	Condition that activates the automation rule.
A_IsActive	TEXT	—	No	Indicates whether the automation rule is active or inactive.
D_Id	INTEGER	FK	No	References the smart device associated with the automation rule.

Foreign Key: `D_Id → Devices(D_Id)`

3.7.8 Notifications

Attribute	Data Type	Key	Nullable	Description
N_Id	INTEGER	PK	No	Unique identifier of the notification.
N_Message	TEXT	—	No	Detailed notification message.
N_DateTime	TEXT	—	No	Date and time when the notification was generated.
N_Type	TEXT	—	No	Type or category of the notification.
N_IsRead	TEXT	—	No	Indicates whether the notification has been read.
U_Id	INTEGER	FK	No	References the user receiving the notification.

Foreign Key: `U_Id → Users(U_Id)`

3.7.9 ActivityLogs

Attribute	Data Type	Key	Nullable	Description
AL_Id	INTEGER	PK	No	Unique identifier of the activity log.
AL_Action	TEXT	—	No	Type of activity performed by the user.
AL_Description	TEXT	—	Yes	Detailed description of the activity.
AL_DateTime	TEXT	—	No	Date and time when the activity occurred.

U_Id	INTEGER	FK	No	References the user who performed the activity.
D_Id	INTEGER	FK	No	References the device related to the activity.

Foreign Keys:

- `U_Id → Users(U_Id)`
- `D_Id → Devices(D_Id)`

3.7.10 EnergyConsumptions

Attribute	Data Type	Key	Nullable	Description
E_Id	INTEGER	PK	No	Unique identifier of the energy consumption record.
E_PowerUsed	REAL	—	No	Amount of power consumed by the device.
E_DateTime	TEXT	—	No	Date and time when energy consumption was recorded.
D_Id	INTEGER	FK	No	References the smart device.

Foreign Key: `D_Id → Devices(D_Id)`

3.8 SQL Implementation

A. Table Creation Queries:

1. Users

```
CREATE TABLE Users (
   U_Id INTEGER PRIMARY KEY AUTOINCREMENT,
   U_Mail TEXT NOT NULL UNIQUE,
   U_Name TEXT NOT NULL,
   U_Role TEXT NOT NULL,
   U_Pass TEXT NOT NULL,
   U_Acc_Status TEXT NOT NULL
);
```

2. Rooms

```
CREATE TABLE Rooms (
    R_Id INTEGER PRIMARY KEY AUTOINCREMENT,
    R_Type TEXT NOT NULL,
    R_Name TEXT NOT NULL
);
```

3. Devices

```
CREATE TABLE Devices (
    D_Id INTEGER PRIMARY KEY AUTOINCREMENT,
    D_Name TEXT NOT NULL,
    D_InstallationDate TEXT NOT NULL,
    D_Type TEXT NOT NULL,
    D_PowerState TEXT NOT NULL,
    D_Status TEXT NOT NULL,
    R_Id INTEGER NOT NULL,

    FOREIGN KEY (R_Id) REFERENCES Rooms(R_Id)
);
```

4. Permissions

```
CREATE TABLE Permissions (
    P_Id INTEGER PRIMARY KEY AUTOINCREMENT,
    P_Status TEXT NOT NULL,
    P_AccessType TEXT NOT NULL,
    P_GrantedDate TEXT NOT NULL,
    P_ExpireDate TEXT,
    U_Id INTEGER NOT NULL,
    D_Id INTEGER NOT NULL,
```

FOREIGN KEY (U_Id) REFERENCES Users(U_Id),
    FOREIGN KEY (D_Id) REFERENCES Devices(D_Id)
);

5. UserPhoneNumbers

```
CREATE TABLE UserPhoneNumbers (
    U_Id INTEGER NOT NULL,
    U_Number TEXT NOT NULL,

    PRIMARY KEY (U_Id, U_Number),
    FOREIGN KEY (U_Id) REFERENCES Users(U_Id)
);
```

6. UserRooms

```
CREATE TABLE UserRooms (
    U_Id INTEGER NOT NULL,
    R_Id INTEGER NOT NULL,

    PRIMARY KEY (U_Id, R_Id),
    FOREIGN KEY (U_Id) REFERENCES Users(U_Id),
    FOREIGN KEY (R_Id) REFERENCES Rooms(R_Id)
);
```

7. AutomationRules

```
CREATE TABLE AutomationRules (
    A_Id INTEGER PRIMARY KEY AUTOINCREMENT,
    A_Action TEXT NOT NULL,
    A_Name TEXT NOT NULL,
    A_TriggerCondition TEXT NOT NULL,
    A_IsActive TEXT NOT NULL,
```

```
    D_Id INTEGER NOT NULL,

    FOREIGN KEY (D_Id) REFERENCES Devices(D_Id)
);
```

8. Notifications

```
CREATE TABLE Notifications (
    N_Id INTEGER PRIMARY KEY AUTOINCREMENT,
    N_Message TEXT NOT NULL,
    N_DateTime TEXT NOT NULL,
    N_Type TEXT NOT NULL,
    N_IsRead TEXT NOT NULL,
    U_Id INTEGER NOT NULL,

    FOREIGN KEY (U_Id) REFERENCES Users(U_Id)
);
```

9. ActivityLogs

```
CREATE TABLE ActivityLogs (
    AL_Id INTEGER PRIMARY KEY AUTOINCREMENT,
    AL_Action TEXT NOT NULL,
    AL_Description TEXT NOT NULL,
    AL_DateTime TEXT NOT NULL,
    U_Id INTEGER NOT NULL,
    D_Id INTEGER,

    FOREIGN KEY (U_Id) REFERENCES Users(U_Id),
    FOREIGN KEY (D_Id) REFERENCES Devices(D_Id)
);
```

10. EnergyConsumptions

```
CREATE TABLE EnergyConsumptions (
    E_Id INTEGER PRIMARY KEY AUTOINCREMENT,
    E_PowerUsed REAL NOT NULL,
    E_DateTime TEXT NOT NULL,
    D_Id INTEGER NOT NULL,

    FOREIGN KEY (D_Id) REFERENCES Devices(D_Id)
);
```

B. Data Insertion Queries:

1 Users Data

```
INSERT INTO Users
(U_Mail, U_Name, U_Role, U_Pass, U_Acc_Status)
VALUES
('sajid@gmail.com', 'Md Sajid Arafat', 'Homeowner', 'pass123', 'Active'),
('rahim@gmail.com', 'Abdur Rahim', 'Family Member', 'rahim123', 'Active'),
('karim@gmail.com', 'Abdul Karim', 'Guest', 'karim123', 'Active'),
('technician@gmail.com', 'Rakib Hasan', 'Technician', 'tech123', 'Active');
```

2 Rooms Data

```
INSERT INTO Rooms
(R_Type, R_Name)
VALUES
('Bedroom', 'Master Bedroom'),
('Living Room', 'Living Room'),
('Kitchen', 'Main Kitchen'),
```

('Bathroom', 'Common Bathroom');

3 Devices Data

INSERT INTO Devices

(D_Name, D_InstallationDate, D_Type, D_PowerState, D_Status, R_Id)

VALUES

('Smart LED Bulb', '2026-07-31', 'Light', 'ON', 'Active', 1),

('Smart Fan', '2026-07-31', 'Fan', 'OFF', 'Active', 1),

('Smart TV', '2026-07-31', 'Television', 'OFF', 'Active', 2),

('Smart Refrigerator', '2026-07-31', 'Appliance', 'ON', 'Active', 3),

('Smart Camera', '2026-07-31', 'Security Camera', 'ON', 'Active', 2);

4 Permissions Data

INSERT INTO Permissions

(P_Status, P_AccessType, P_GrantedDate, P_ExpireDate, U_Id, D_Id)

VALUES

('Active', 'Full Access', '2026-07-31', NULL, 1, 1),

('Active', 'Full Access', '2026-07-31', NULL, 1, 2),

('Active', 'Full Access', '2026-07-31', NULL, 1, 3),

('Active', 'Control', '2026-07-31', '2026-12-31', 2, 2),

('Active', 'Read Only', '2026-07-31', '2026-08-15', 3, 3);

5 UserPhoneNumbers Data

INSERT INTO UserPhoneNumbers

(U_Id, U_Number)

VALUES

(1, '01711111111'),

(1, '01811111111'),

(2, '01922222222'),

(3, '01633333333'),

(4, '01544444444');

6 UserRooms Data

INSERT INTO UserRooms

(U_Id, R_Id)

VALUES

(1, 1),

(1, 2),

(1, 3),

(1, 4),

(2, 1),

(2, 2),

(3, 2),

(4, 3);

7 ActivityLogs Data

INSERT INTO ActivityLogs

(AL_Action, AL_Description, AL_DateTime, U_Id, D_Id)

VALUES

('Login', 'User logged into the system', '2026-07-31 09:00:00', 1, NULL),

('Device Control', 'Turned ON Smart LED Bulb', '2026-07-31 09:05:00', 1, 1),

('Permission Granted', 'Family member received device access', '2026-07-31 09:10:00', 1, 2),

('Device Control', 'Turned OFF Smart Fan', '2026-07-31 09:20:00', 2, 2),

('Logout', 'User logged out', '2026-07-31 09:30:00', 1, NULL);

8 Notifications Data

INSERT INTO Notifications

(N_Message, N_DateTime, N_Type, N_IsRead, U_Id)

VALUES

('Smart Bulb turned ON', '2026-07-31 09:05:00', 'Device', 'No', 1),

('New Permission Granted', '2026-07-31 09:10:00', 'Permission', 'Yes', 2),

('Guest Access Expiring Soon', '2026-08-10 10:00:00', 'Warning', 'No', 3);

9 AutomationRules Data

INSERT INTO AutomationRules

(A_Action, A_Name, A_TriggerCondition, A_IsActive, D_Id)

VALUES

('Turn ON', 'Morning Lights', '06:00 AM', 'Yes', 1),

('Turn OFF', 'Night Fan', '11:00 PM', 'Yes', 2),

('Turn ON', 'Security Camera', 'Motion Detected', 'Yes', 5);

10 EnergyConsumptions Data

INSERT INTO EnergyConsumptions

(E_PowerUsed, E_DateTime, D_Id)

VALUES

(12.50, '2026-07-31 08:00:00', 1),

(18.20, '2026-07-31 09:00:00', 2),

(55.75, '2026-07-31 10:00:00', 3),

(102.40, '2026-07-31 11:00:00', 4),

(8.90, '2026-07-31 12:00:00', 5);

4. Conclusion:

The database design of the Smart Home Management System provides a structured and reliable foundation for managing the system's core data. The database consists of normalized and interconnected tables that support user management, room and device management, access permissions, automation rules, notifications, activity tracking, and energy consumption monitoring. Primary keys and foreign keys are used to maintain data integrity and establish relationships among the entities. The database has been implemented using SQLite 3 with appropriate constraints and sample data. The designed structure is also flexible enough to support future system expansion and integration with the ASP.NET Core Web API and mobile application.